

MOTOR (Steven)

Rpm

→ C++

→ Arduino Library

→ The computer is wobbling on one foot trying to keep itself as balanced as possible, if it overcorrects it can reconnect

Steven will make the physical presentation

Main.cpp

Hall effect sensor

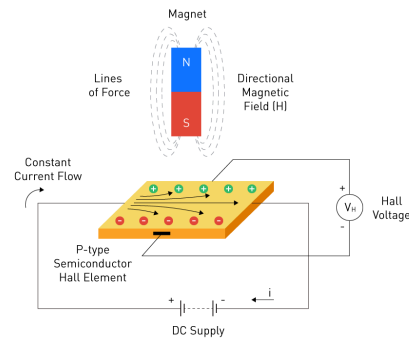
→ A Hall effect sensor is a transducer (a device that converts energy from one form to another) that varies its output voltage in response to an external magnetic field, acting as a non-contact, durable, and highly accurate magnetic sensor. Used for position detection, speed measurement, and proximity switching, these devices operate by detecting the Lorentz force on charge carriers in a semiconductor, producing a measurable

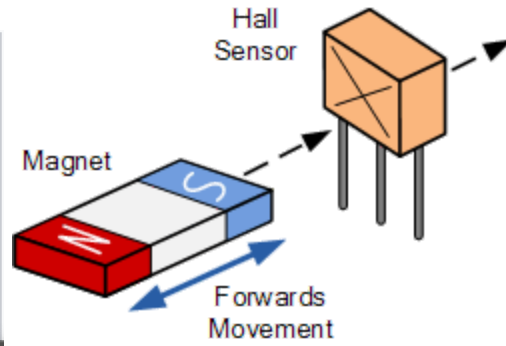
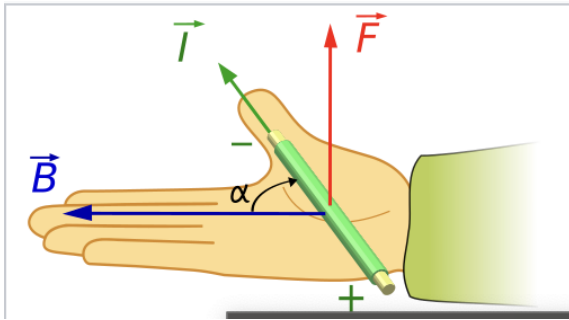
(taken from the book)

→ The Lorentz force (electric force + magnetic force) is the total electromagnetic force exerted on a charged particle moving with velocity through both an electric field and a magnetic field.

$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$$

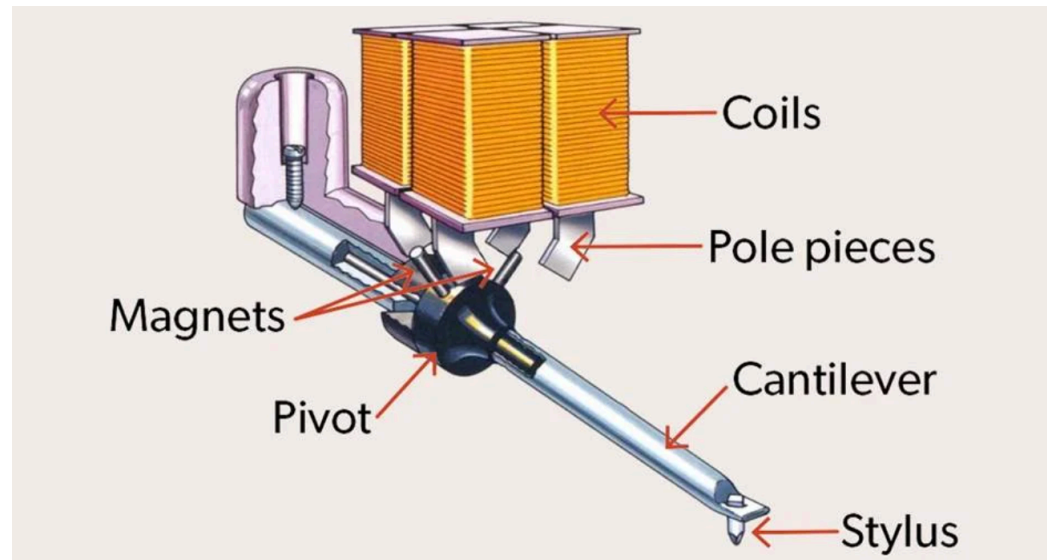
Electric force *Magnetic force*





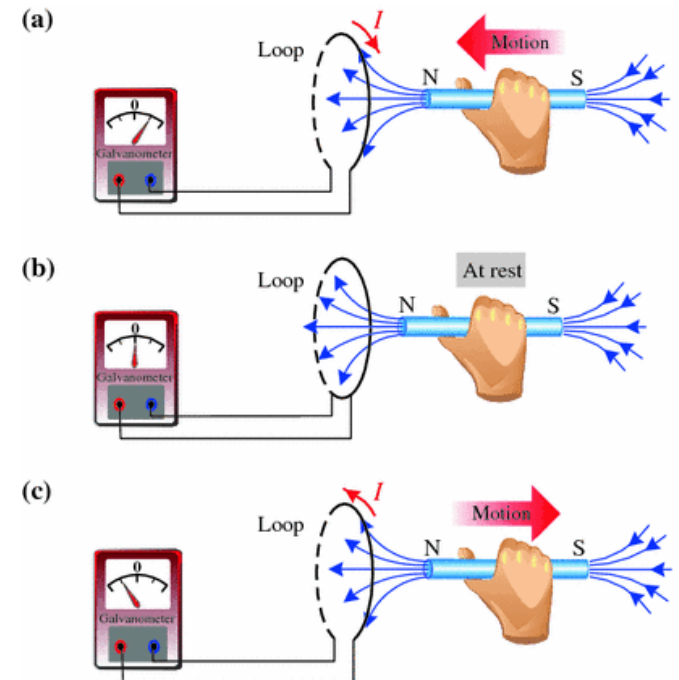
→ Electric force is parallel to the electric field. Magnetic force is perpendicular to the velocity and magnetic field.
Hall sensor is the transducer

TONE ARM



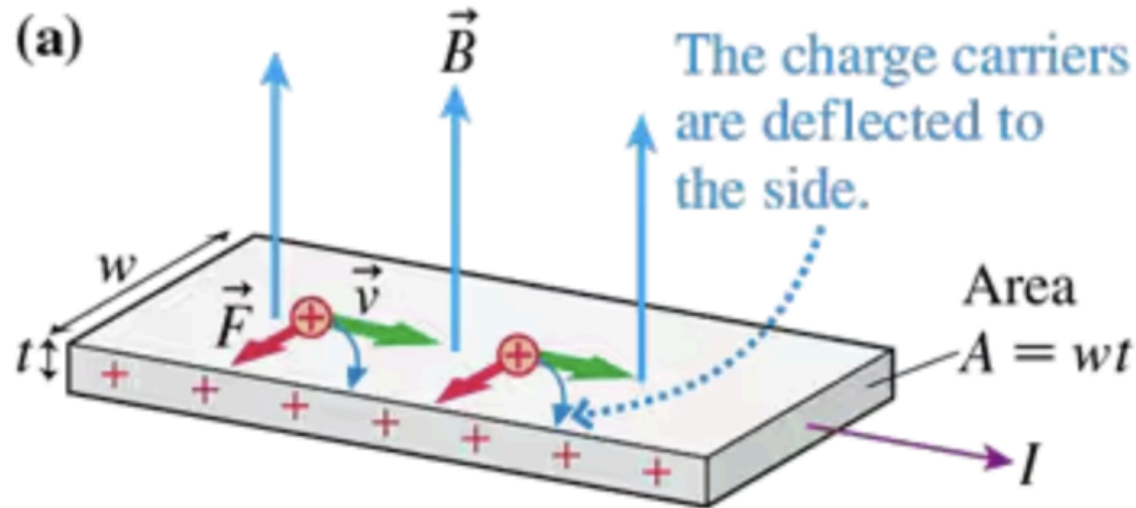
Faraday's law

- Mechanical to electrical
- The vibrations created by the stylus interacting with the disc are converted into electrical signals.
- Magnetic flux is a measurement of the total magnetic field passing through a given area.
- Faraday's Law describes when magnetic flux changes in a conductor that creates force. Generators use this concept to create energy through spinning a turbine.

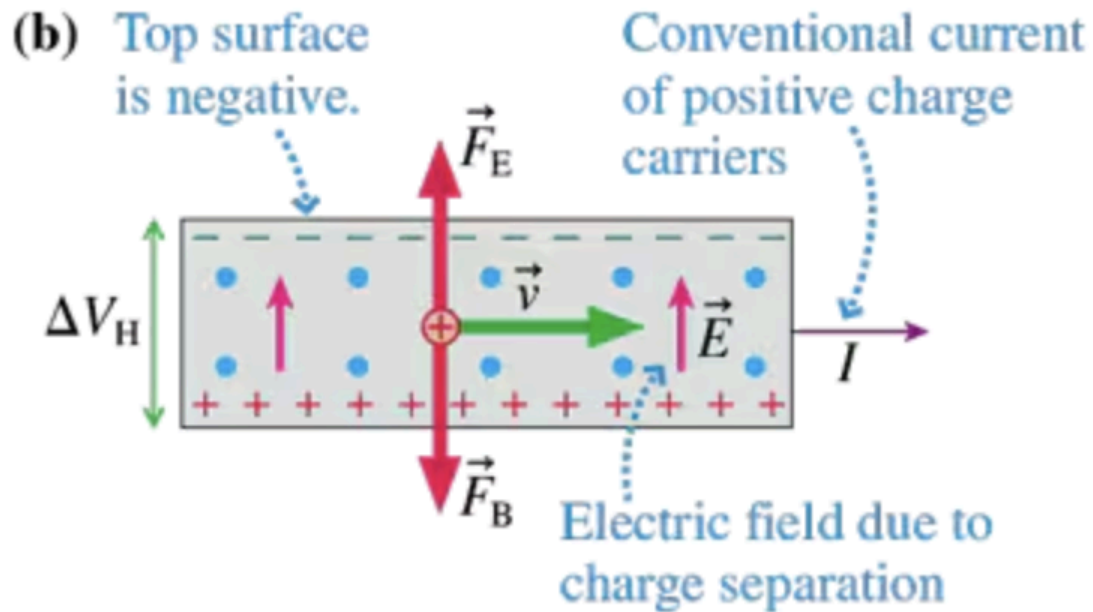


Hall Effect Sensor

- The hall effect allows us to collect information about charge carriers in a conductor. It helps us measure magnetic field strengths.
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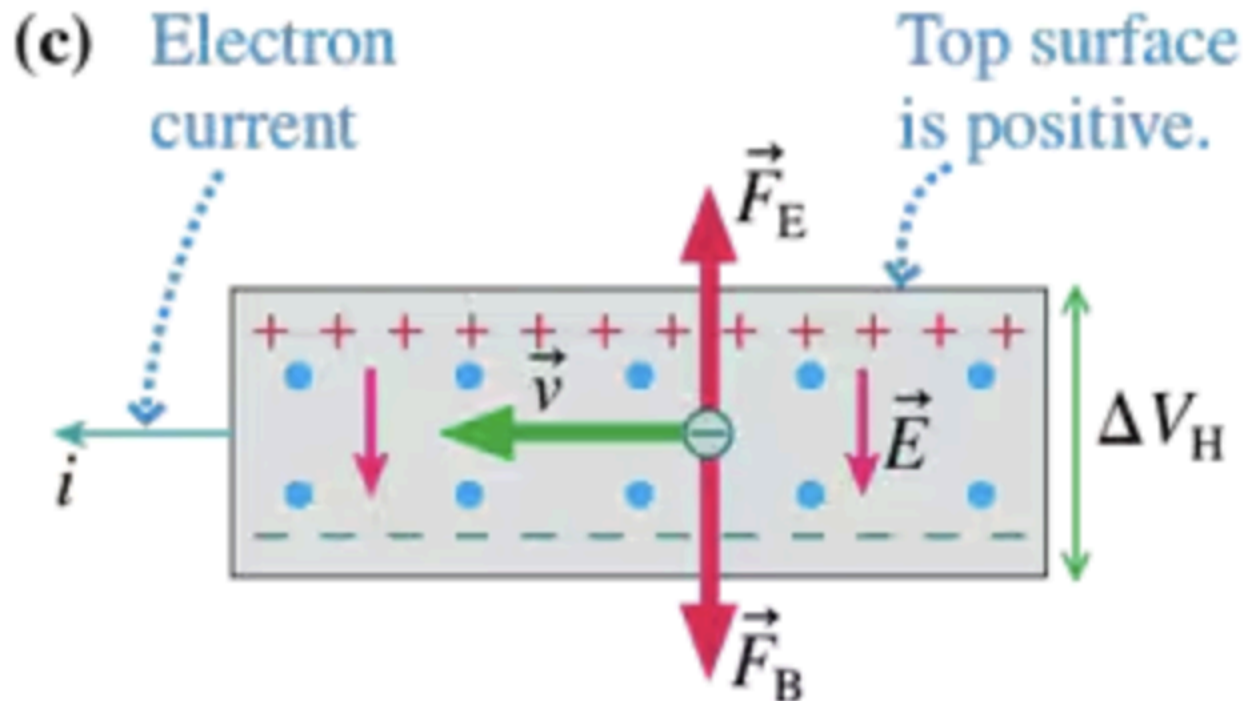


- In this image, the magnetic field is perpendicular to the conductor. The positive signs represent charges. The conductor carries current across itself. In this specific scenario, it is necessary to know whether the charge carrier is positive or negative.



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→ The positive charge carriers are moving in the direction of the current, which pushes them to the bottom surface of the conductor, which leaves an excess negative charge on the top. Using the right hand rule, the magnetic force is pushing the positive charge carriers to the bottom of the conductor



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- If we were to use our left hand, the electron current goes in the opposite direction of the conventional current, and the magnetic force on the electron goes down.
- The sign of the excess charge on the bottom of the surface is the same as the sign of the charge carriers. The bottom surface is negative when the conductor is a metal, which is evidence that the charge carriers in metals are electrons.
- Electrons are deflected towards the bottom of the surface when the current begins to flow, but it cannot continue indefinitely. As excess charge accumulates on the top and bottom surface, it acts

like the charge on the plates of a capacitor. A potential difference is created between the two surfaces. This electric field increases until the upward electric force on the charge carriers balances the magnetic force that goes in the opposite direction.

$$F_B = F_E$$

→

- The steady-state potential difference between the two surfaces of the conductor is the hall voltage. Hall voltage is larger for poor conductors that have smaller charge-carrier densities.
- This information is useful in creating sensors in electrical devices.
- Magnetic field is perpendicular to a conducting strip that has a current. Charge carriers are pulled to opposite edges, creating the Hall Voltage.

Measurable difference to the two sides

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-
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1. Faraday's Law

a.

2. Hall Effect Sensor

3. Lorentz Law + equation

$$\vec{F} = q \underset{\substack{\text{Electric} \\ \text{force}}}{\vec{E}} + q \vec{v} \times \underset{\substack{\text{Magnetic} \\ \text{force}}}{\vec{B}}$$

a.

4.

Motor

Electrical energy to mechanical energy

- Stator Magnets: Provides a constant magnetic field, on the outer body of the motor.
- Rotorcoils: Coiled copper wires that become an electromagnet.
- Commutator: A split ring that flips the current direction to keep the rotor spinning.
- Brushes: Spring-loaded carbon or graphite blocks that "brush" against the commutator to transfer electrical power from the stationary input to the moving rotor.

